Type MS · When is a binomial expansion valid?

Answer 1 of 3 · Question 7 part b · 1 mark

2023 JUN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

7(a)	$\left(1+\frac{x}{3}\right)^{-3}=\left[1+(-3)\left(\frac{x}{3}\right)+\frac{(-3)(-3-1)}{2!}\left(\frac{x}{3}\right)^2+\frac{(-3)(-3-1)(-3-2)}{3!}\left(\frac{x}{3}\right)^3\ldots\right]$ $=1-x+\frac{2}{3}x^2-\frac{10}{27}x^3$	M1A1 A1 [3]
(b)	$-3<x<3$ or $\lvert x\rvert<3$	B1 [1]
(c)	$\left((3+x)^{-3}=3^{-3}\times\left(1+\frac{x}{3}\right)^{-3}=\frac{1}{27}\times\left(1+\frac{x}{3}\right)^{-3}\right)$ $P=\frac{1}{27}$, $Q=\frac{1}{3}$	 B1, B1 [2]
(d)	$\frac{(1+4x)}{(3+x)^3}=(1+4x)\times\frac{1}{27}\times\left(1-x+\frac{2}{3}x^2\right)$ $=\frac{1}{27}\times\left(1-x+\frac{2}{3}x^2+4x-4x^2+\ldots.\right)$ $=\frac{1}{27}\left(1+3x-\frac{10x^2}{3}\right)$ or $\frac{1}{27}+\frac{x}{9}-\frac{10x^2}{81}$	 M1 A1 [2]
(e)	$\int_0^{0.2}\left(\frac{1}{27}+\frac{x}{9}-\frac{10x^2}{81}\right)\mathrm{d}x=\left[\frac{x}{27}+\frac{x^2}{18}-\frac{10x^3}{243}\right]_0^{0.2}$ $=\left(\frac{0.2}{27}+\frac{0.2^2}{18}-\frac{10\times0.2^3}{243}\right)-[0]$ $=0.0093004$	M1 M1 A1 [3]
		Total 11 marks

Type MS · When is a binomial expansion valid?

Answer 1 of 3 · Question 7 part b · 1 mark

2023 JUN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For applying a correct binomial expansion in unsimplified form. Minimum required: • The expansion begins with 1 • The next term is correct • The powers of $\frac{x}{3}$ must be correct eg $\left(\frac{x}{3}\right)^2$ • The denominators are correct. Do not allow missing brackets unless recovered later – this is a general point of marking. Ignore any terms with powers higher than 3.
	A1	Following M1 (this is a general point of marking, A marks can only follow M marks), all conditions above met, must see $1 - x$ and at least the term in x^2 or x^3 correct and simplified. Ignore any terms with powers higher than 3.
	A1	A fully correct and simplified expansion. Ignore any terms with powers higher than 3..
(b)	B1	For the correct validity.
(c)	B1	For the correct value of P or Q explicitly written or embedded in $\frac{1}{27}\times\left(1+\frac{x}{3}\right)^{-3}$
	B1	For the correct values of P and Q explicitly written or embedded in $\frac{1}{27}\times\left(1+\frac{x}{3}\right)^{-3}$
(d)	M1	For attempting to multiply their expansion, which must be of the form "P"× (their expansion from part (a), with a minimum of 3 terms, by $(1 + 4x)$. An attempt must include 3 correctly multiplied out terms of their expansion before simplification. The "P" may remain factorised. Ignore any terms which to powers of x higher than 2.
	A1	For the correct expansion in either form shown. Allow equivalent coefficients. Ignore powers of x higher than 2.
(e)	M1	For an attempt to integrate their expression from part (d), provided it has at least one constant term and at least one algebraic term. See general guidance, no power of x must decrease.
	M1	For substituting in the value of 0.2 into a changed expression and subtracting the correct way. Must see the explicit substitution of 0.2 at least once, if the final answer is not correct. Can be implied if final answer correct. Substitution of 0 does not need to be seen.
	A1	For the value of 0.0093004 [The calculator value is 0.0093316] Accept the value of $\frac{113}{12150}$ if seen.

Type MS · When is a binomial expansion valid?

Answer 1 of 3 · Question 7 part b · 1 mark

2023 JUN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · When is a binomial expansion valid?

Answer 2 of 3 · Question 9 part b · 1 mark

2023 JUN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

9(a)	$(1+2x)^{-\frac{1}{3}} = 1+\left(-\frac{1}{3}\right)(2x)+\frac{\left(-\frac{1}{3}\right)\left(-\frac{4}{3}\right)(2x)^2}{2!}+\frac{\left(-\frac{1}{3}\right)\left(-\frac{4}{3}\right)\left(-\frac{7}{3}\right)(2x)^3}{3!}+\ldots$ $=1-\frac{2x}{3}+\frac{8x^2}{9}-\frac{112x^3}{81}$	M1 A1A1 [3]
(b)	$-\frac{1}{2} < x \leqslant \frac{1}{2}$ accept $-\frac{1}{2} < x < \frac{1}{2}$	B1 [1]
(c)	$f(x) = (2+kx^2)\left(1-\frac{2x}{3}+\frac{8x^2}{9}-\frac{112x^3}{81}\right)$ $=2-\frac{4x}{3}+\frac{16}{9}x^2+kx^2-\frac{224}{81}x^3-\frac{2k}{3}x^3$	M1 M1A1 [3]
(d)	$-\frac{224}{81}-\frac{2k}{3}=-\frac{8}{3} \Rightarrow k=-\frac{4}{27}$	M1A1 [2]
(e)	$\int_{0.1}^{0.2}\left(2-\frac{4x}{3}+\frac{44x^2}{27}-\frac{8}{3}x^3\right)dx=\left[2x-\frac{4x^2}{2\times3}+\frac{44x^3}{3\times27}-\frac{8x^4}{4\times3}\right]_{0.1}^{0.2}$ $=\left(2\times0.2-\frac{2\times0.2^2}{3}+\frac{44\times0.2^3}{81}-\frac{2}{3}\times0.2^4\right)-\left(2\times0.1-\frac{2\times0.1^2}{3}+\frac{44\times0.1^3}{81}-\frac{2}{3}\times0.1^4\right)$ $=0.1828$ NB: Calculator value is 0.18301744	B1FT M1A1ft M1 A1 [5]
		Total 14 marks

Type MS · When is a binomial expansion valid?

Answer 2 of 3 · Question 9 part b · 1 mark

2023 JUN · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For an attempt at binomial expansion. • The first term is 1 • The denominators are correct. • The powers of x are correct.
	A1	For at least one algebraic term correct and simplified.
	A1	For a fully correct and simplified expansion.
(b)	B1	For the correct range. Accept $\|x\| < \frac{1}{2}$ Do not accept $-\frac{1}{2} \leq x \leq \frac{1}{2}$
(c)	M1	For showing the intent to multiply their expansion by $\left(2+kx^2\right)$
	M1	For expanding the two brackets up to terms in x^3. Ignore terms in higher powers.
	A1	For the correct expansion. This does not have to be simplified. Must be in ascending powers, but accept $\frac{16}{9}x^2$ and kx^2 in either order and accept $-\frac{224}{81}x^3$ and $-\frac{2k}{3}x^3$ in either order. $f(x) = 2 - \frac{4x}{3} + x^2\left(\frac{16}{9}+k\right) + x^3\left(-\frac{224}{81}-\frac{2k}{3}\right)$ ISW errors in collection of coefficients once correct expansion in ascending powers seen.
(d)	M1	For setting their coefficient of x^3 equal to $-\frac{8}{3}$
	A1	For the correct value of k
(e)	B1FT	For substituting in their k into coefficient for x^2 in their expansion from (c) and having the coefficient of x^3 as $-\frac{8}{3}$. If the coefficient of x^2 is not correct then the substitution must be seen to award this mark. Note: this is an M mark in epen.
	M1	For an attempt to integrate their expansion which must have 4 terms. At least two powers of x to increase by 1 (including $c \rightarrow cx$). No power of x to reduce.
	A1ft	For a correctly integrated expression follow through their 4 term polynomial. Simplified integrated expression is $2x - \frac{2x^2}{3} + \frac{44x^3}{81} - \frac{2}{3}x^4$
	M1	For substituting in 0.2 and 0.1 the correct way around in their integrated expression. Expression should not have been differentiated and needs to have changed. Not dependent on first M. Must show the substitution if their integral or their limits are incorrect. Correct awrt 0.1828 following correct integral with correct limits implies this mark.
	A1	For the correct value of awrt 0.1828 NB: The calculator value is 0.18301744

Type MS · When is a binomial expansion valid?

Answer 2 of 3 · Question 9 part b · 1 mark

2023 JUN · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · When is a binomial expansion valid?

Answer 3 of 3 · Question 2 part b · 1 mark

2024 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

2(a)	$\frac{2}{\sqrt{1+3x}} = 2(1+3x)^{\left(-\frac{1}{2}\right)}$	B1
	$\{2\}\left(1+\left(-\frac{1}{2}\right)(3x)+\frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)(3x)^2}{2!}+\frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)(3x)^3}{3!}+\ldots\right)$	M1A1
	$2-3x+\frac{27x^2}{4}-\frac{135x^3}{8}+\ldots$	A1 [4]
(b)	$-\frac{1}{3}<x<\frac{1}{3}$ (Accept $\lvert x\rvert<\frac{1}{3}$)	B1 [1]
		Total 5 marks

Part	Mark	Notes
(a)	B1	Correct simplification of the given expression. This mark may be implied by a correct expansion.
	M1	Attempts at the binomial expansion $(1+3x)^{\pm\frac{1}{2}}$ or $2(1+3x)^{\pm\frac{1}{2}}$ For an attempt at the binomial expansion. • The first term is 1 or 2 • The powers of $3x$ are correct in all terms, e.g. $(3x)^2$ • The correct denominators are used, 2! and 3! oe
	A1	Allow this mark for at least 1 correct algebraic term, correctly simplified, from $-3x+\frac{27x^2}{4}-\frac{135x^3}{8}$ or $-\frac{3}{2}x+\frac{27x^2}{8}-\frac{135x^3}{16}$
	A1	Correct simplified expansion in ascending order, coefficients must be in simplest **fractions**. (Ignore extra terms with powers > 3) Do not isw
(b)	B1	For the correct range of values of x. $-\frac{1}{3}<x<\frac{1}{3}$ or $\lvert x\rvert<\frac{1}{3}$, do not accept $\lvert 3x\rvert<1$

ALT for (a) Uses **Maclaurin's expansion (If seen send to review)**

$$f(x) = f(0) + f'(0)x + \frac{f''(x)}{2!}x^2 + \frac{f'''(x)}{3!}x^3 + \ldots$$

B1 Correct simplification of the given expression.

M1 Achieves $f'(x) = P(1+3x)^{-\frac{3}{2}}, f''(x) = Q(1+3x)^{-\frac{5}{2}}, f'''(x) = R(1+3x)^{-\frac{7}{2}}, P, Q, R \neq 0$ and attempts to find the values of *f'(0), f''(0)* and *f'''(0)*

A1 Correct unsimplified expansion OR at least 2 correct simplified terms

A1 Fully correct simplified expansion. (Ignore extra terms with powers higher than 3)

For reference, the correct derivatives are:

$$f'(x) = -3(1+3x)^{-\frac{3}{2}}, f''(x) = \frac{27}{2}(1+3x)^{-\frac{5}{2}}, f'''(x) = \frac{-405}{4}(1+3x)^{-\frac{7}{2}}$$

Type MS · When is a binomial expansion valid?

Answer 3 of 3 · Question 2 part b · 1 mark

2024 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question	Scheme	Marks